

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

April-May 2013

EC311 VLSI Technology & Design

Date: 03/05/2013

Time: 10:00 am to 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain Regularity and Modularity concepts for VLSI design. [02]
(b) What do you mean by Channel Hot Electron effect? [02]
(c) Discuss the effect of Substrate Bias (Body effect) with example. [03]
- Q - 2 (a) Explain depletion region for nMOS transistor and Derive the depletion region depth X_d equation for the same. [05]
(b) Explain CMOS ring oscillator. [05]
(c) Consider a CMOS inverter with the following parameters: [04]
nMOS: $V_{T0,n} = 0.6V$ $\mu_n C_{ox} = 60 \mu A/V^2$
pMOS: $V_{T0,p} = -0.8V$ $\mu_p C_{ox} = 20 \mu A/V^2$
Also $V_{DD} = 3V$, $\lambda = 0$.
(i) Determine the (W/L) ratios of the nMOS and pMOS transistor such that the switching threshold is $V_{th} = 1.5V$.
(ii) Calculate the V_{OH} and V_{OL} values.

OR

- Q - 2 (a) Derive the expression for switching power dissipation of CMOS inverter. [06]
(b) Determine the aspect ratio W/L, for a resistive load inverter with $R = 1 k\Omega$ and $V_{OL} = 0.6V$. The enhancement-type nMOS driver transistor has the following parameters: [03]
 $V_{DD} = 5.0V$, $V_{T0} = 1.0V$, $\gamma = 0.2 V^{1/2}$, $\lambda = 0$, $\mu_n C_{ox} = 22.0 \mu A/V^2$

- (c) Measured voltage and current data for MOSFET are given below. Determine the type [05]
of the device and calculate the parameters k_n , V_{T0} , and γ . Assume Fermi potential is
-0.3V

$V_{GS}(V)$	$V_{DS}(V)$	$V_{SB}(V)$	$I_D(\mu A)$
3	3	0	97
4	4	0	235
5	5	0	433
3	3	3	59
4	4	3	173
5	5	3	347

- Q - 3 (a) What is the advantage of JK latch over SR latch? Draw and Explain working of [07]
CMOS JK latch in detail.
- (b) Explain the Band diagram of the MOS structure, at surface inversion and also derive [07]
the expression of threshold voltage.

OR

- Q - 3 (a) Describe various device isolation techniques. Explain bird's beak problem in LOCOS [07]
isolation technique.
- (b) Consider the depletion-load nMOS NOR2 gate with following parameters: $\mu_n C_{ox} = 25$ [07]
 $\mu A/V^2$, $V_{T0,driver} = 1$ V, $V_{T0,load} = -3$ V, $\gamma = 0.4$ V^{1/2}, and $|2\phi_F| = 0.6$ V. The transistor
dimensions are given as $(W/L)_A = 2$, $(W/L)_B = 2$, and $(W/L)_{load} = 1/3$. The power
supply voltage is $V_{DD} = 5$ V.
Calculate the output voltage levels for all four valid input voltage combinations.

SECTION - II

- Q - 4 (a) Realize the following Boolean function: [02]
(i) $Z = [A(D+E)+BC]'$ using nMOS complex logic gate.
(ii) Two input Multiplexers using transmission gates.
- (b) Draw the circuit diagram of D-type latch using transmission gate. [02]
- (c) What is the minimum power supply required on CMOS inverter so that it can operate [03]
properly?

- Q - 5 (a) Prove that, if the area occupied by the depletion-type load transistor is negligible then [06]
two-input NAND gate will occupy approximately four-times the area occupied by the
equivalent inverter that has the same static characteristics. Neglect substrate-bias
effect.
- (b) State the requirement of voltage bootstrapping. Draw the Dynamic bootstrapping [08]
arrangement and explain it in detail.

OR

- Q - 5 (a) Discuss the various techniques available for on chip clock generation and distribution [08]
for successful high-speed VLSI design.
- (b) Draw and Explain the operation of enhancement-load nMOS dynamic shift register. [06]
- Q - 6 (a) Explain Build-in self test technique in detail. [06]
- (b) Explain VTC of Depletion Load nMOS inverter with power and area considerations. [04]
- (c) Discuss the impact of different VLSI design styles upon the design cycle time and the [04]
achievable circuit performances.

OR

- Q - 6 (a) Explain Ad-hoc testable design techniques in detail. [06]
- (b) Derive the expression $(W/L)_p \cong 2.5 (W/L)_n$ for symmetric CMOS inverter. [04]
- (c) Draw and explain equivalent circuit of CMOS inverter in the case of high to low [04]
transition of the output.
